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23CSE211- DESIGN AND ANALYSIS OF ALGORITHMS

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Class : CSE-B

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WEEK-7

Program-1: To implement the Greedy algorithm for the Job Sequencing with Deadlines problem and determine the optimal sequence of jobs that maximizes total profit.

Consider a set of 14 jobs, where each job requires one unit of time for completion. Each job has an associated profit and deadline. A job must be completed on or before its deadline to earn the profit.

The profits of the jobs are:

{22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 27, 19, 11}

The deadlines of the jobs are:

{3, 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 2, 14, 1}

Code:

```
#include <stdio.h>
#define MAX 50

struct Job {
    int id;
    int profit;
    int deadline;
};

int main() {

    printf("Job Sequencing with Deadlines (Greedy Method)\n");

    int n = 14;    // given number of jobs
    int i, j;

    struct Job jobs[MAX];

    // Given profits and deadlines
    int profits[14] = {22,19,29,28,30,21,27,25,24,26,14,27,19,11};
    int deadlines[14] = {3,3,8,6,7,5,10,4,6,12,13,2,14,1};

    // Assign values to job structure
    for(i = 0; i < n; i++) {
        jobs[i].id = i + 1;
        jobs[i].profit = profits[i];
        jobs[i].deadline = deadlines[i];
    }
```

```

// ♦ Sort jobs by profit (descending)
for(i = 0; i < n-1; i++) {
    for(j = 0; j < n-i-1; j++) {
        if(jobs[j].profit < jobs[j+1].profit) {
            struct Job temp = jobs[j];
            jobs[j] = jobs[j+1];
            jobs[j+1] = temp;
        }
    }
}

// ♦ Find maximum deadline
int maxDeadline = 0;
for(i = 0; i < n; i++) {
    if(jobs[i].deadline > maxDeadline)
        maxDeadline = jobs[i].deadline;
}

int slot[MAX];
int totalProfit = 0;

// Initialize slots as empty
for(i = 0; i <= maxDeadline; i++)
    slot[i] = -1;

```

```

// ♦ Assign jobs to slots
for(i = 0; i < n; i++) {
    for(j = jobs[i].deadline; j > 0; j--) {
        if(slot[j] == -1) {
            slot[j] = jobs[i].id;
            totalProfit += jobs[i].profit;
            break;
        }
    }
}

// ♦ Output schedule
printf("\nTime Slot\tJob\n");
for(i = 1; i <= maxDeadline; i++) {
    if(slot[i] != -1)
        printf("%d\t\tJ%d\n", i, slot[i]);
    else
        printf("%d\t\t-\n", i);
}

printf("\nMaximum Profit = %d\n", totalProfit);

return 0;
}

```

Output:

```
lahari@DESKTOP-JIGL8HI:~$ ./job_seq
Job Sequencing with Deadlines (Greedy Method)

Time Slot      Job
1              J6
2              J12
3              J1
4              J8
5              J9
6              J4
7              J5
8              J3
9              --
10             J7
11             --
12             J10
13             J11
14             J13

Maximum Profit = 292
```

Time Complexity: $O(N^2)$:

The program takes $O(N^2)$ time because it uses bubble sort to order jobs by profit and a nested loop to assign jobs to available time slots in the worst case.

Space Complexity: $O(N)$:

The program requires $O(N)$ space to store the jobs and the time slot array, which are proportional to the number of jobs.

WORKING

JOB SEQUENCING

To implement the greedy algorithm for the job sequencing with deadlines and determine the optimal sequence of jobs that maximises the total profit.

Consider a set of 14 jobs, where each job requires one unit of time for completion. Each job has an associated profit and deadline.

The profits of jobs are

{ 22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 27, 19, 17 }

The deadlines of the jobs are:

{ 3, 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 2, 14, 1 }

Step 1: Arrange the jobs in descending order of profit

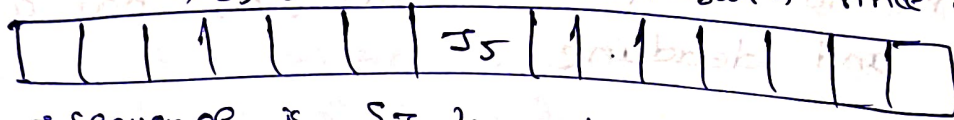
JOB	PROFIT	DEADLINE
J ₅	30	7
J ₃	29	8
J ₄	28	6
J ₇	27	10
J ₁₂	27	2
J ₁₀	26	12
J ₈	25	4
J ₉	24	6
J ₁	22	3
J ₆	21	5
J ₂	19	9
J ₁₃	19	14
J ₁₁	14	18
J ₁₄	11	1

Since the maximum value of deadline is 14, fourteen, time slots are created

Step 1: Assign jobs at the optimal time slots.

01) Add $J_5 (30, 7)$

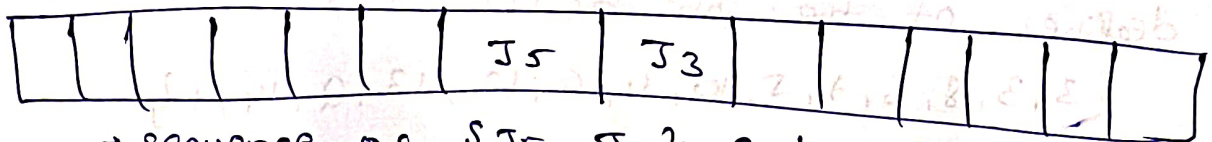
→ J_5 can be added in slot 7 since it is empty



→ sequence is $\{J_5\}$ and the profit is 30

02) Add $J_3 (29, 8)$

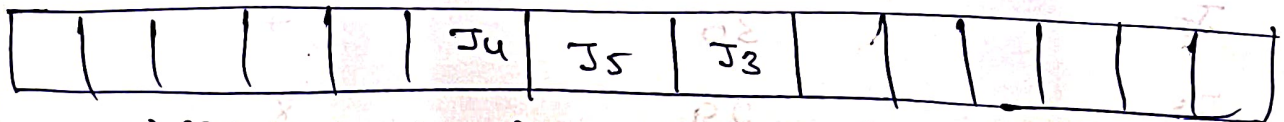
→ J_3 can be added in slot 8 since it is empty



→ sequence is $\{J_5, J_3\}$ and the profit is 59.

03) Add $J_4 (28, 6)$

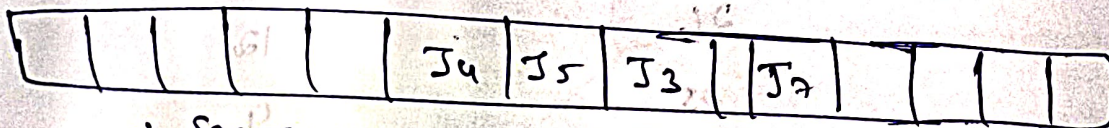
→ J_4 can be added in slot 6 since it is empty



→ sequence is $\{J_5, J_3, J_4\}$ and profit is 87.

04) Add $J_7 (27, 10)$

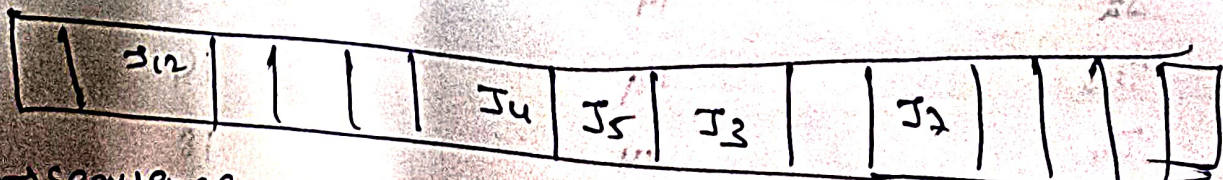
→ J_7 can be added in slot 10 since it is empty



→ sequence is $\{J_5, J_3, J_4, J_7, J_{12}\}$ and profit is 119

05) Add $J_{12} (27, 2)$

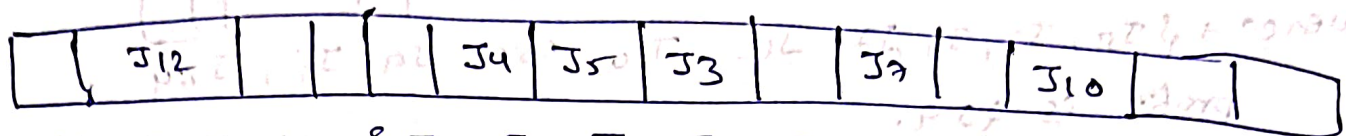
→ J_{12} can be added in slot 2 since this is empty



→ sequence is $\{J_5, J_3, J_4, J_7, J_{12}\}$ and profit is 141

66) Add J_{10} (26, 4)

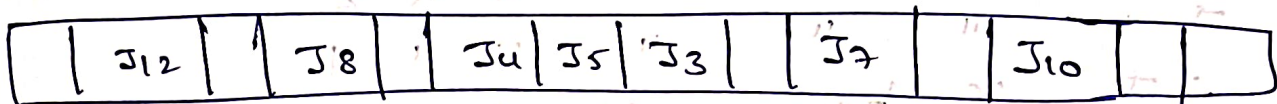
→ J_{10} can be added in slot 4 since it is empty



→ Sequence is $\{J_5, J_3, J_4, J_7, J_{12}, J_{10}\}$ and profit is 162.

07) Add J_8 (25, 4)

→ J_8 can be added in slot 4 since it is empty

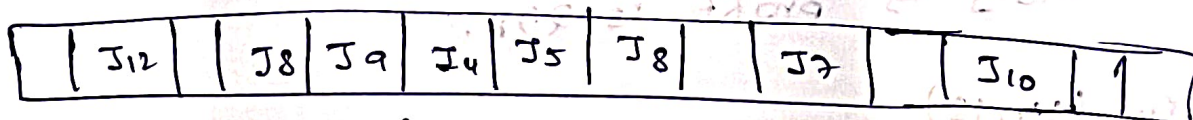


→ Sequence is $\{J_5, J_3, J_4, J_7, J_{12}, J_{10}, J_8\}$, and the profit is 192

08) Add J_9 (24, 6)

→ J_9 can't be added in slot 6 since it is occupied

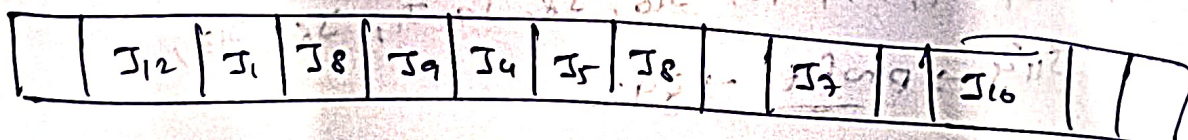
→ But it can be added to slot -5 since it is empty



→ Sequence is $\{J_8, J_3, J_4, J_7, J_{12}, J_{10}, J_8, J_9\}$ profit = 218

09) Add J_1 (22, 3)

→ J_1 can be added in slot 3 since it is empty



→ Sequence is $\{J_5, J_3, J_4, J_7, J_{12}, J_{10}, J_8, J_9, J_1\}$ and profit is 236.

10) Add J_6 (21, 5)

→ J_6 can't be added in slot 5

→ " " " " " " 4

→ " " " " " " 3

→ " " " " " " 2

→ " can't be added to slot -1 since it has

J ₆	J ₁₂	J ₁	J ₈	J ₉	J ₄	J ₅	J ₃		J ₇		J ₁₀		
----------------	-----------------	----------------	----------------	----------------	----------------	----------------	----------------	--	----------------	--	-----------------	--	--

→ sequence → {J₅, J₃, J₄, J₇, J₁₂, J₁₀, J₈, J₉, J₁, J₆}
profit is 259.

11) Add J₂ (19, 3)

→ J₂ can't be added in slot - 3

→ " " " " " " 2

→ " " " " " " 1

∴ J₂ is being skipped

12) Add J₁₃ (19, 14)

→ J₁₃ can be added in slot 14 since it is empty

J ₆	J ₁₂	J ₁	J ₈	J ₉	J ₄	J ₅	J ₃		J ₇	J ₁₀	J ₁₃
----------------	-----------------	----------------	----------------	----------------	----------------	----------------	----------------	--	----------------	-----------------	-----------------

→ sequence: {J₅, J₃, J₄, J₇, J₁₂, J₁₀, J₈, J₉, J₁, J₆, J₁₃} → profit is 278.

13) Add J₁₁ (14, 13)

→ J₁₁ can be added in slot 13 since it is empty

J ₆	J ₁₂	J ₁	J ₈	J ₉	J ₄	J ₅	J ₃		J ₇		J ₁₀	J ₁₁	J ₁₃
----------------	-----------------	----------------	----------------	----------------	----------------	----------------	----------------	--	----------------	--	-----------------	-----------------	-----------------

sequence → {J₅, J₃, J₄, J₇, J₁₂, J₁₀, J₈, J₉, J₁, J₆, J₁₃, J₁₁} → profit → 292

14) Try adding J₁₄ (11, 1)

→ J₁₄ can't be added since slot 1 is occupied.

FINAL JOB SEQUENCE

J ₆	J ₁₂	J ₁	J ₈	J ₉	J ₄	J ₅	J ₃		J ₇		J ₁₀	J ₁₁	J ₁₃
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Using the greedy job sequencing algorithm, the optimal sequence of jobs yields a maximum profit: 292.